

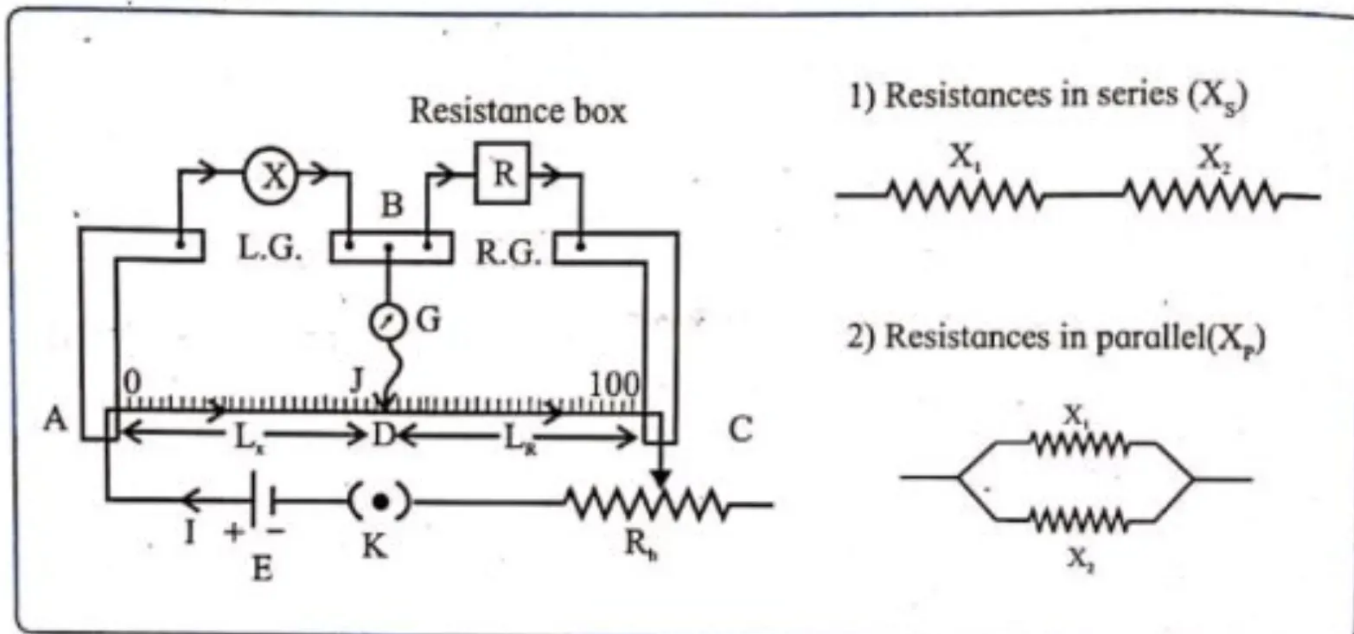
EXPERIMENT NO. 8

LAW OF RESISTANCES USING METER BRIDGE

Aim : To verify the laws of combination (series and parallel) of resistances using a meter bridge.

Apparatus : Meter bridge, two resistances, galvanometer, resistance box, rheostat, cell, jockey, plug key, connecting wires.

Circuit diagram :



Formulae :

- 1) $X = R \left(\frac{L_x}{L_R} \right)$ Where, L_x and L_R are balancing lengths corresponding to X and R.
- 2) For series combination, $X_s = X_1 + X_2$
- 3) For parallel combination, $X_p = \frac{X_1 \times X_2}{X_1 + X_2}$

Procedure :

1. Arrange and connect the circuit for series combination.
2. Take some suitable value of resistance from the resistance box and then close the plug key.
3. Touch the jockey at point A and C of the meter bridge wire, so that the galvanometer will show deflection on either side of zero.
4. By touching the jockey at various points along the wire AC, locate the point where galvanometer shows zero deflection. This point is called as 'NULL POINT'
5. Measure the length L_x and L_R as shown in the figure. Hence, calculate the value of X_s .
6. By changing the resistance in the resistance box, take THREE sets of observations in the same way.
7. Repeat the experiment by interchanging the positions of X_s and R. Hence, calculate, the value of X_s .
8. Repeat the experiment by connecting X_1 and X_2 in parallel, and hence calculate the value of X_p .

Observations :

Resistance (X_1) = $\frac{11.34}{\dots\dots\dots} \Omega$

Resistance (X_2) = $\frac{2.395}{\dots\dots\dots} \Omega$

1) For series combination :

Resistance Position	Obs. No.	Resistance from Resistance Box (Ω) R	L_x (cm)	L_R (cm)	$X_s = R \left(\frac{L_x}{L_R} \right) \Omega$
Series in left gap	1	5	81.5	18.5	22.02
	2	10	67	33	20.30
	3	15	52	48	16.25
Series in right gap	1	5	24	76	1.57
	2	10	38	62	6.12
	3	15	52	48	16.25

Mean $X_s = \frac{13.75}{\dots\dots\dots} \Omega$

2) For parallel combination :

Resistance Position	Obs. No.	Resistance from Resistance Box (Ω) R	L_x (cm)	L_R (cm)	$X_p = R \left(\frac{L_x}{L_R} \right) \Omega$
Parallel in left gap	1	1	55	45	1.22
	2	2	49.5	50.5	1.96
	3	3	44	56	2.35
Parallel in right gap	1	1	43.5	56.5	0.76
	2	2	49	51	1.92
	3	3	54	46	3.52

Mean $X_p = \frac{1.95}{\dots\dots\dots} \Omega$

3) For series combination:

Obs. No	Left gap			Right gap		
	1	2	3	1	2	3
Log R = a	0.6990	1	1.1761	0.6990	1	1.1761
Log $l_x = b$	1.9112	1.8261	1.7160	1.3802	1.5798	1.7160
c = a + b	2.6102	2.8261	2.8921	2.0792	2.5798	2.8921
Log $L_R = d$	1.2672	1.5185	1.6812	1.8808	1.7924	1.6812
c - d = e	1.3430	1.3076	1.2109	0.1984	0.7874	1.2109
Antilog e = X_s	22.02	20.30	16.25	1.57	6.12	16.25

2) For parallel combination :

Obs. No	Left gap			Right gap		
	1	2	3	1	2	3
Log R = a	0	0.3010	0.4771	0	0.3010	0.4771
Log $L_x = b$	1.7404	1.6946	1.6435	1.6385	1.6902	1.7324
c = a + b	1.7404	1.9956	2.1206	1.6385	1.9912	2.2095
Log $L_R = d$	1.6532	1.7033	1.7482	1.7520	1.7076	1.6628
c - d = e	0.0852	0.2923	0.3724	0.8865	0.2836	0.5467
Antilog e = X_p	1.22	1.96	2.35	0.76	1.92	3.52

Calculations :

$$1) X_s = R \left(\frac{L_s}{L_R} \right)$$

$$\text{Mean } X_s = 13.75 \Omega$$

(experimental value)

$$2) X_p = R \left(\frac{L_s}{L_R} \right)$$

$$\text{Mean } X_p = 1.95 \Omega$$

(experimental value)

Result :

1) In series

a) Experimental value of $X_s = 13.75 \Omega$

b) Theoretical value of $X_s = X_1 + X_2 = 13.75 \Omega$

2) In parallel

a) Experimental value of $X_p = 1.95 \Omega$

b) Theoretical value of $X_p = \frac{X_1 \times X_2}{X_1 + X_2} = 1.95 \Omega$

Hence laws of series and parallel is verified.

Precautions :

1. All the connections and plugs in the resistance box should be tight.
2. Null point is obtained in the middle ($\frac{1}{3}$)rd portion of the metre bridge wire.
3. Do not slide the jockey on the wire.
4. Remove the plug key after completion of experiment.

Questions

1. What is a balance point in meter bridge?

The point when Galvanometer shows Null Deflection is called as Balanced point.

2. Resistance of a wire is $x \Omega$, The wire is stretched to triple its length, then the resistance becomes ?

The resistance of wire is inversely proportional to its area and is directly proportional to its length. When length is tripled then resistance will also become three times.

Remark and sign of teacher :